Genetics

- Genetics is a branch of biology dealing with inheritance and variation of characters from parents to offspring.
- **Inheritance** -Process by which characters are passed on from parent to progeny
- **Variation-** Degree by which the progeny differs from its parents

NOTE –

- Human knew from as early as 8000 – 1000 B.C. that one of the causes of variation was hidden in sexual reproduction.
- They exploited the variations that were naturally present in the wild populations of plants and animals to selectively breed and select for organisms that possessed desirable characters.
- For example, through artificial selection and domestication from ancestral wild cows, we have well known Indian breeds, e.g., **Sahiwal cows** in Punjab.

Definitions of various terms used in genetics

- **Gene –** it is defined as 'the functional units of inheritance. They contain information that is required to express a particular trait, in an organism'. (They are the units which Mendel refer to as **factor** in his breeding experiments).

- **Alleles –** genes which code for a pair of contrasting trait are known as alleles, they are slightly different forms of the same gene.

(OR)

They are 'a pair of genes present at a specific locus on homologous chromosomes but, determining the alternate expression of the same character'. Example – Round and wrinkled, tall and short, yellow and green are all two alternate expression of a particular character referred to as alleles.

- **Locus** (pl. Loci) **–** it is 'a specific region or point on the chromosome at which a gene or its alleles reside'.

- **Phenotype –** it is 'the external appearance of an organism'. It is expressed in words like tall, dwarf etc.

- **Genotype –** it is 'the genetic constitution of an organism'. It is expressed by symbols like TT, Tt, tt, RR, Rr, rr etc.

- **Homozygous -** it is 'a condition where the paired genes are similar on homologous chromosomes for a character'. Eg. TT (tall), tt (dwarf)

- **Heterozygous –** it is 'a condition where the paired genes are dissimilar on homologous chromosomes for a character. Eg. Tt (tall)

- **Hybrid –** it is 'an offspring of two parents differing in one or more characters'. Eg. Homozygous tall (TT) crossed with homozygous dwarf (tt) produce tall plants (Tt) which are hybrids.

- **Dominant gene –** it is 'a gene which express itself in the presence of a contrasting gene'.

- **Recessive gene –** it is 'a gene which fails to express itself in the presence of a dominant gene'.

- **Dominant character –** it is 'a character which is completely expressed even in the presence of a recessive allele'.

- **Monohybrid –** it is 'a plant or an animal obtained by crossing two homozygous individuals differing in one character'.
- **Dihybrid -** it is 'a plant or animal obtained by crossing two homozygous individuals differing in two pairs of contrasting characters'.
- **Back cross –** a cross between a F_1 individual and either of its parents is called a backcross
- **Test cross -** it is a backcross between F1 or F2 hybrid (unknown genotype) with a homozygous recessive parent.
- **Out cross –** it is a backcross between F1 or F2 hybrid (unknown genotype) with a homozygous dominant parent.

Mendel's Experiment

- **Gregor Johann Mendel** known as the **father of genetics** proposed the **laws of inheritance**.
- He conducted hybridization experiments on **garden peas** (*Pisum sativum)* for **seven years** (1856 – 1863).
- He was the first person who used the statistical analysis and mathematical logic in biology.
- Large sampling size gave credibility to his collected data.
- Mendel investigated characters in the garden pea plant that were manifested as opposite traits. Example – Tall and Drawf.
- Mendel conducted artificial pollination/cross pollination/hybridization experiment using several true breeding pea lines.
- **True breeding pea lines** – are one that having undergone self-pollination for several generations and shows stable inheritance and expression of traits for several generations.
- Mendel selected 14 true breeding pea lines as 7 pairs, which were similar except for one character with contrasting traits.
- He worked on the following **seven** characters of garden pea:

Character	Stem Height	Seed Shape	Seed Color	Pod Shape	Pod Color	Flower Color	Flower Position
Dominant trait	Tall	Round	Yellow	Inflated (Full)	Green	Violet	Axial
Recessive trait	Dwarf	Wrinkled	Green	Constricted	Yellow	White	Terminal

Figure: Seven pairs of contrasting traits in pea plants studied by Mendel

Mendel choose the pea plants (*Pisum sativum*) **for his work because,**

- They were easy to cultivate.
- Require minimum space and time.
- The plants possess a number of contrasting characters.
- They have a short life span.
- The flowers are naturally self-pollinating.
- It was possible to conduct cross pollination by transferring pollen grains form one flower to another.
- All pea plants produce a large number of seeds.

Inheritance of One Gene

- Mendel performed artificial cross pollination (hybridisation) on true breeding pea lines to obtain first hybrid generation known as the first filial progeny or F_1.

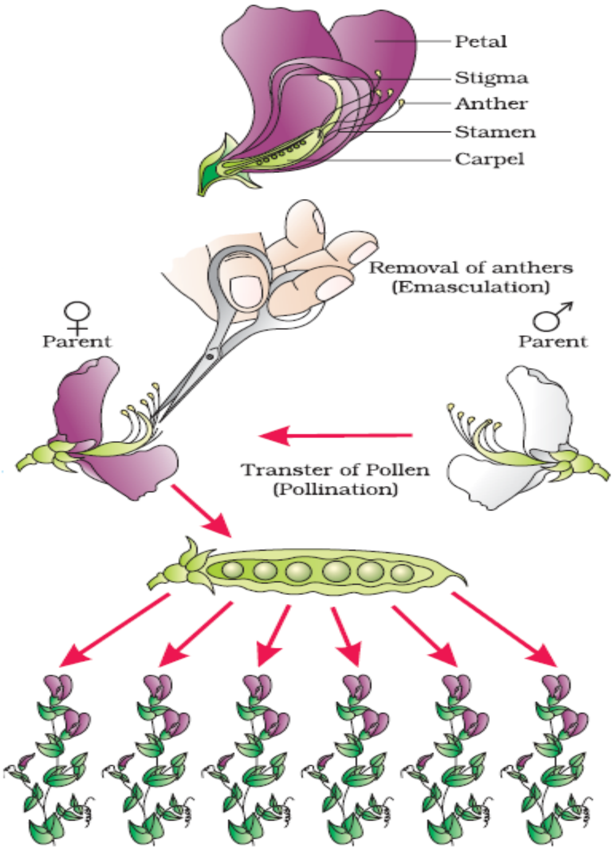


Figure: Steps in making a cross in pea

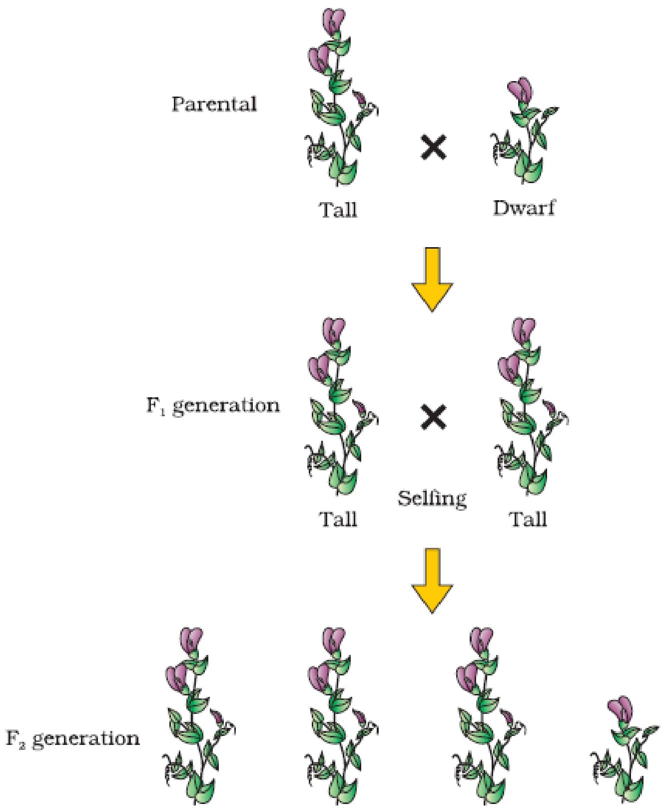
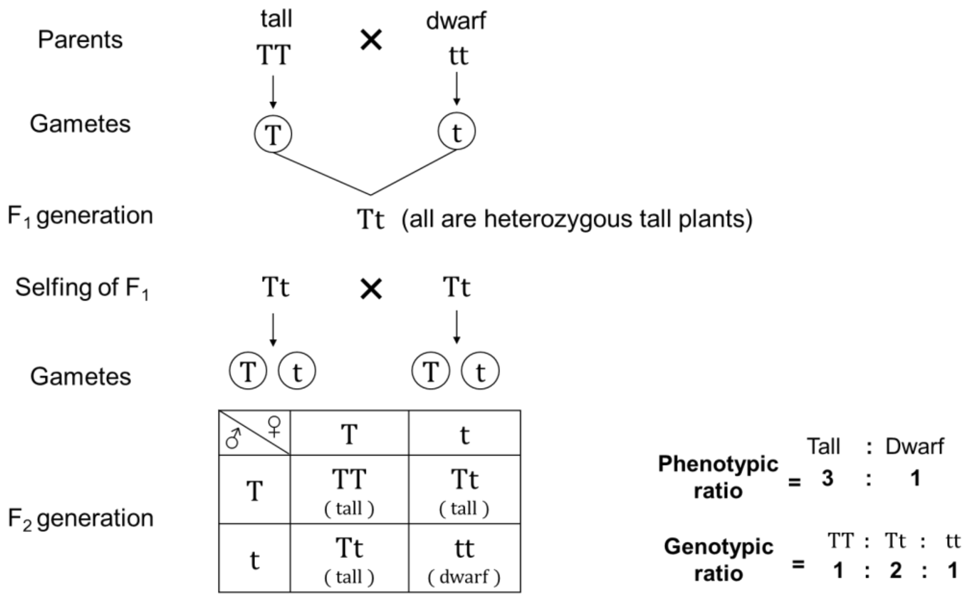


Figure: Diagrammatic representation of monohybrid cross

- After hybridisation, the F_1 generation so obtained resembled only one of its parents (say, all tall; no dwarf).
- When plants from F_1 generation were self-pollinated, the second filial progeny or F_2 generation was obtained.
- Revival of unexpressed trait (dwarf) was observed in some F_2 progeny. Both traits, tall and dwarf, were expressed in F_2 in ratio 3:1.
- Mendel proposed that something is being passed unchanged from generation to generation. He called these things as **'factors'** (presently called genes).
- Factors contain and carry hereditary information.
- **Alleles** – slightly different form of same factor.
- Two alleles code for a pair of contrasting traits. (e.g., tall and dwarf)

Monohybrid Cross

- Cross that considers only a single character (e.g., height of the plant) is called **monohybrid cross**.



- TT, Tt, and tt are **genotypes** while the traits, tall and dwarf, are **phenotypes**.
- 'T' stands for tall trait while 't' stands for dwarf trait.
- Even if a single 'T' is present in the genotype, phenotype is 'tall'. When 'T' and't' are present together, 'T' dominates and suppresses the expression of 't'. Therefore, 'T' (for tallness) is **dominant trait** while 't' (for dwarfness) is **recessive trait.**
- 'TT' and 'tt' are **homozygous** while 'Tt' is **heterozygous**.
- From the cross, it can be found that alleles of parental pair separate or segregate from each other and only one allele is transmitted to the gamete.
- Gametes of 'TT' will have only 'T' alleles; gametes of 'tt' will have only 't' alleles, but gametes of 'Tt' will have both 'T' and 't' alleles.

Mendel's Laws of Inheritance

- Based on his experiments, Mendel proposed three laws or principles of inheritance:
  i. Law of Dominance
  ii. Law of Segregation
  iii. Law of Independent Assortment

- Law of dominance and law of segregation are based on monohybrid cross while law of independent assortment is based on dihybrid cross.

Mendel's First Law / Law of Dominance

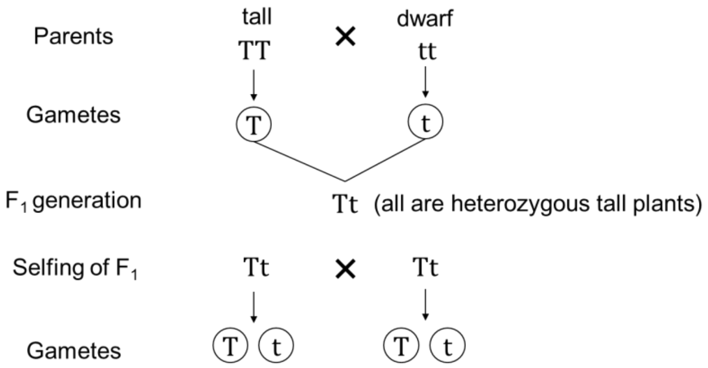
- According to this law, characters are controlled by discrete units called factors, which occur in pairs with one member of the pair dominating over the other in a dissimilar pair.
- This law explains expression of only one of the parental character in F_1 generation and expression of both in F_2 generation.

Mendel's Second Law / Law of Segregation / Law of purity of gametes

- This law states that the two alleles of a pair segregate or separate during gamete formation such that a gamete receives only one of the two factors.
- In homozygous parents, all gametes produced are similar; while in heterozygous parents, two kinds of gametes are produced in equal proportions.

Punnett square / Checker board

- Graphical representation to calculate the probability of all possible genotypes of offsprings in a genetic cross.
- It was developed by British geneticist, **Reginald C, Punnett.**
- Possible gametes are written on two sides, usually at top row and left columns and combinations are represented in boxes.



F_2 generation

♂ \ ♀	T	t
T	TT (tall)	Tt (tall)
t	Tt (tall)	tt (dwarf)

Phenotypic ratio = Tall : Dwarf = 3 : 1

Genotypic ratio = TT : Tt : tt = 1 : 2 : 1

- With the help of Punnet square, genotypic ratio in F_2 generation can be found. From the above given Punnet square, it is evident that genotypic ratio TT: Tt: tt is 1:2:1.
- The ratio 1:2:1 or $\frac{1}{4}:\frac{1}{2}:\frac{1}{4}$ of TT: Tt: tt can be derived from **binomial expression (*ax* + *by*)²**.

Where

'**a**' and '**b**' are the gamete bearing genes **T** or **t** in equal frequency of **½** .

'**x**' is the dominant allele.

'**y**' is the recessive allele.

Hence, the expression can be expanded as

$$\left(\frac{1}{2}T+\frac{1}{2}t\right)^2 = \left(\frac{1}{2}T+\frac{1}{2}t\right) \times \left(\frac{1}{2}T+\frac{1}{2}t\right)$$

$$= \frac{1}{4}TT+\frac{1}{4}Tt+\frac{1}{4}Tt+\frac{1}{4}tt$$

$$= \frac{1}{4}TT+2\left(\frac{1}{4}Tt\right)+\frac{1}{4}tt$$

$$= \frac{1}{4}TT+\frac{1}{2}Tt+\frac{1}{4}tt$$

Test Cross

- Cross between F_1or F_2 progeny and its homozygous recessive parent is called **test cross**.
- **Significance -** This cross determines whether the dominant character is coming from homozygous dominant genotype or heterozygous genotype. (e.g., tallness coming from TT or Tt)
- When TT is crossed with tt, we obtain all Tt (tall) individuals in the progeny. Whereas when Tt is crossed with tt, we obtain Tt (tall) and tt (dwarf) individuals in the progeny in the ratio **1 : 1** and in Dihybrid test cross the ratio will be **1 : 1 : 1 : 1**
- Therefore, if tallness is coming from TT, then we obtain all tall progenies in test cross. We obtain both tall and dwarf varieties in test cross, if tallness is coming from Tt.

Case 1

(Unknown genotype) (Homozygous recessive)

TT X tt

♂ \ ♀	t	t
T	Tt (tall)	Tt (tall)
T	Tt (tall)	Tt (tall)

Result: All plants are tall

Interpretation: Unknown plant is homozygous dominant

Case 2

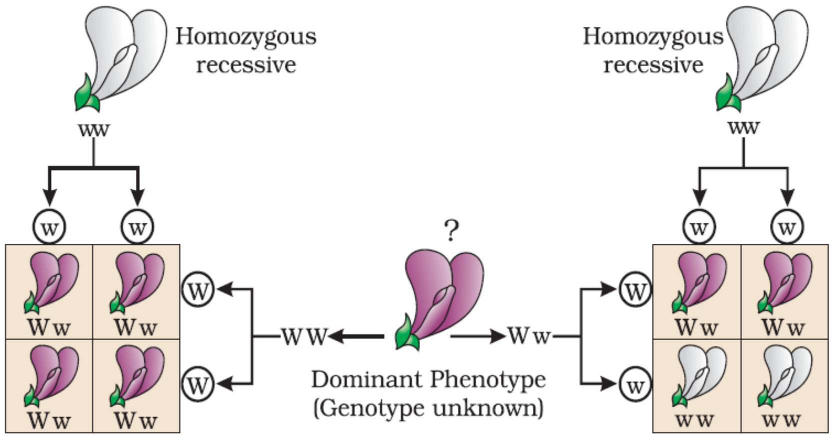
(Unknown genotype) (Homozygous recessive)

Tt X tt

♂ \ ♀	t	t
T	Tt (tall)	Tt (tall)
t	tt (dwarf)	tt (dwarf)

Result: Half of the plants are tall
Half of the plants are dwarf

Interpretation: Unknown plant is heterozygous



Result: All flowers are violet | Half of the flowers are violet and half of the flowers are white.

Interpretation: Unknown flower is homozyous dominant | Unknown flower is heterozygous

Figure: Diagrammatic representation of a test cross

Monohybrid cross in animals – Coat colour in guinea pig

- In guinea pigs (*Cavia porcellus*), there are two types of coat colour **black coat (BB)** and **white coat (bb).** if we cross a black coated homozygous male (pure breed) with a homozygous white coated female (pure breed) we obtain **hybrids** only black coated animals with a heterozygous genotype **(Bb)** in the F1 generation i.e., each sperm will carry one gene for black and each egg will carry one gene for white. Thus, all the offsprings of this cross are heterozygous i.e., they will carry one of each type of gene. And, since all these heterozygous pigs will be black, we can conclude that black is dominant over white.

Parents: black BB × white bb

Gametes: B b

F_1 generation: Bb (all black coated, but heterozygous monohybrids)

Selfing of F_1: Bb × Bb

Gametes: B b B b

F_2 generation:

♂ \ ♀	B	b
B	BB (black)	Bb (black)
b	Bb (black)	bb (white)

Phenotypic ratio = Black : White **3 : 1**

Genotypic ratio = BB : Bb : bb **1 : 2 : 1**

Incomplete Dominance / Intermediate inheritance - Deviation from Mendelian law

- In incomplete dominance, F_1 generation has a phenotype that does not resemble either of the two parents, but is a mixture of the two.
- Discovered by **Carl Correns** in 1903 in Four O'clock plant.
- Examples – Flower colour in *Antirrhinum majus* (snapdragon/dog flower) and in *Mirabilis jalapa* (Four O' clock plant), Andalusian fowls.

Example 1 - Antirrhinum *majus* (dog flower/snapdragon) & *Mirabilis jalapa* (Four O'clock plant)

Where:

- RR – Red flowers
- rr – White flowers
- Rr – Pink flowers

Parents: red RR × white rr

Gametes: R, r

F_1 generation: Rr (all are pink)

Selfing: Rr × Rr

Gametes: R r, R r

F_2 generation:

♂ \ ♀	R	r
R	RR (red)	Rr (pink)
r	Rr (pink)	rr (white)

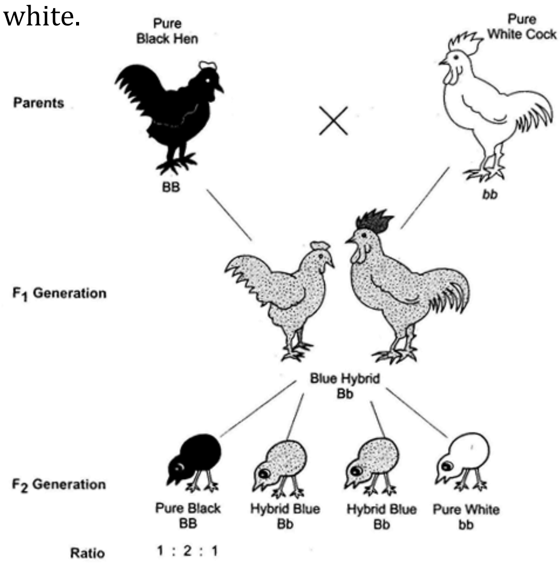
Phenotypic ratio = Red : Pink : White = 1 : 2 : 1

Genotypic ratio = RR : Rr : rr = 1 : 2 : 1

- Here, genotypic ratio remains same as in Mendelian crosses, but phenotypic ratio changes since complete dominance is not shown by R (hence, incomplete dominance).

Example 2 - Andalusian fowls

- Andalusian fowls have two pure forms, black and white. If the two forms are crossed, F1 individuals appear blue coloured due to occurrence of fine alternate black and white stripes on the feathers. Incidentally the blue coloured fowls are favoured as delicacy. F2 generation produces three types of fowls – 1 black : 2 blue : 1 white.



Difference between Dominance and Incomplete Dominance	
Dominance	**Incomplete Dominance**
1. F_1 is similar to the dominant parent	1. F_1 is different from either of the two parents
2. Phenotypic ratio is different from genotypic ration	2. Phenotypic and genotypic ratios are the same
3. In F_1 hybrid, the dominant trait is completely expressed	3. In F_1 hybrid, dominant trait is incompletely expressed.
4. A single dominant allele (e.g., Tt) expresses fully	4. Two alleles (e.g., RR) are required for the expression of complete dominant trait.

Explanation of the concept of dominance.

- Every gene contains the information to express a particular trait.
- A diploid organism there are two copies of a gene, i.e, as a pair of alleles.
- Two alleles need not be identical, as in a heterozygote.
- One of them may be different due to some changes that it has undergone which modifies the information that particular allele contains.
- Suppose a gene produces an enzyme. Now there are two copies of this gene.
- The normal allele produces the normal enzyme that is needed for the transformation of a substrate S
- Then, the modified allele could be responsible for production of –
  - (i) The normal / less efficient enzyme, or
  - (ii) A non-functional enzyme, or
  - (iii) No enzyme at all.
- In the first case, the modified allele is equivalent to the unmodified allele called as **equivalent allele** as it will produce the same phenotype/trait, i.e., result in the transformation of substrate S.
- But, if the allele produces a non-functional enzyme, the phenotype may be effected because, the phenotype /trait will only be dependent on the functioning of the unmodified allele.
- The original gene is said to be dominant while the modified gene is recessive.
- Hence, in the example above the recessive trait is seen due to non-functional enzyme or because no enzyme is produced.

Multiple allelism / Gene polymorphism - Deviation from Mendelian law

- The phenomenon of expression of a single gene in more than two alternative form is called **multiple allelism** or **gene polymorphism.**
- **Multiple allele** is a single gene for a character occurring in more than two alternative form on the same locus of homologous chromosome (or) When more than two alleles control a character, as in human blood groups
- Multiple alleles can be found only when population studies are made.
- Example: ABO blood groups in human beings. ABO blood groups are controlled by gene "*I*". Gene **"*I*"** has three alleles, **"I^A"**, **"I^B"** and **"*i*" (multiple alleles)**

Co-dominance

- The phenomenon of expression of both the alleles in a heterozygote is called **co-dominance**.
- The alleles which do not show dominance-recessive relationship and are able to express themselves independently when present together are called **co-dominant alleles**.
- In co-dominance, the F_1 progeny resembles both the parents.
- Example: "**AB" blood groups in human beings**, **sickle cell anaemia** and **coat colour in cattle.**

Example 1 – ABO Blood Group

- ABO blood groups are controlled by **gene "I"**. Gene "I" has three alleles, "I^A", "I^B" and "i" **(multiple alleles)**.
- As human beings are diploid organism. A person possesses any two of the three alleles.
- "I^A" and "I^B" dominate over "**i**". But with each other, "I^A" and "I^B" are co-dominant.
- "I^A" and "I^B" produce A and B types of sugar (glycoprotein/antigen/agglutinogens), while "i" does not produce any sugar.

NOTE –
- ABO blood group in human beings was discovered by **Karl Landsteiner** in 1901.
- The inheritance of A, B, AB, and O blood group in human beings was discovered by **Felix Bernstein** in 1924.

Allele from Parent 1	Allele from Parent 2	Genotype of offspring	Blood type of offspring
I^A	I^A	I^AI^A	A
I^A	I^B	I^AI^B	AB
I^A	i	I^Ai	A
I^B	I^A	I^AI^B	AB
I^B	I^B	I^BI^B	B
I^B	i	I^Bi	B
i	i	ii	O

- There are 6 genotypes for ABO blood group – I^AI^A, I^Ai, I^BI^B, I^Bi, I^AI^B, ii
- There are 4 phyenotypes for ABO blood group – A, B, AB and O

Exercise on inheritance of ABO blood groups in human beings

1. When the blood group of father and mother are heterozygous for "A" group. What is the blood group of the progeny?

2. When father is homozygous for blood group "A" and mother is heterozygous for blood group "B". What will be the blood group of children?

3. When both father and mother are heterozygous for "A" and "B" blood group. What will be the blood group of the progeny?

4. When the blood group of mother is "AB" and that of father is "O". What will be the blood group of the progeny?

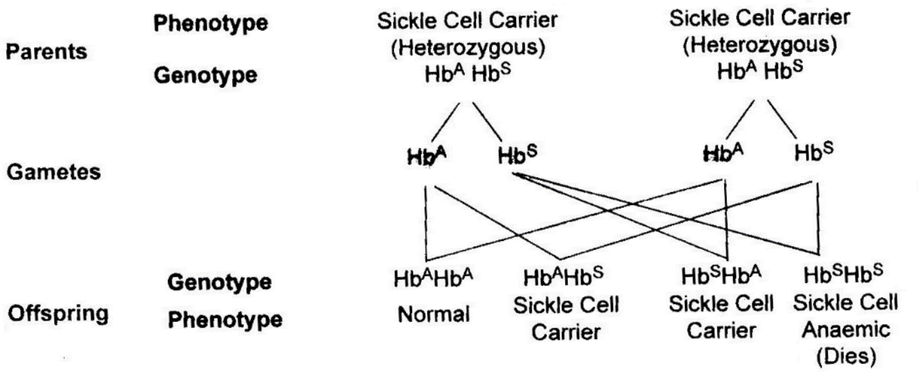
5. A child has blood group "O". If the father has blood group "A" and mother has blood group "B", work out the genotype of the parent and the possible genotype of the other offspring.

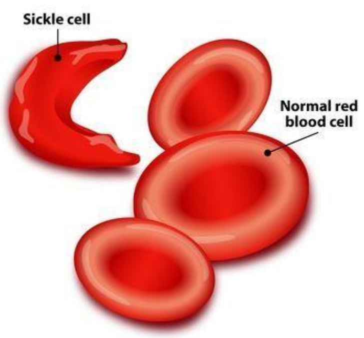
NOTE –

- Parents who are homozygous for **"A"** blood group produce – **"A"** progeny
- Parents who are heterozygous for **"A"** blood group produce – **"A"** and **"O"** progeny
- Parents who are homozygous for **"B"** blood group produce – **"B"** progeny
- Parents who are heterozygous for **"B"** blood group produce – **"B"** and **"O"** progeny
- Parents with **"AB"** blood group produce – **"A"**, **"B"**, & **"AB"** progeny
- Parents with **"O"** blood group produce – **"O"** progeny

Example 2 – Sickle cell anaemia

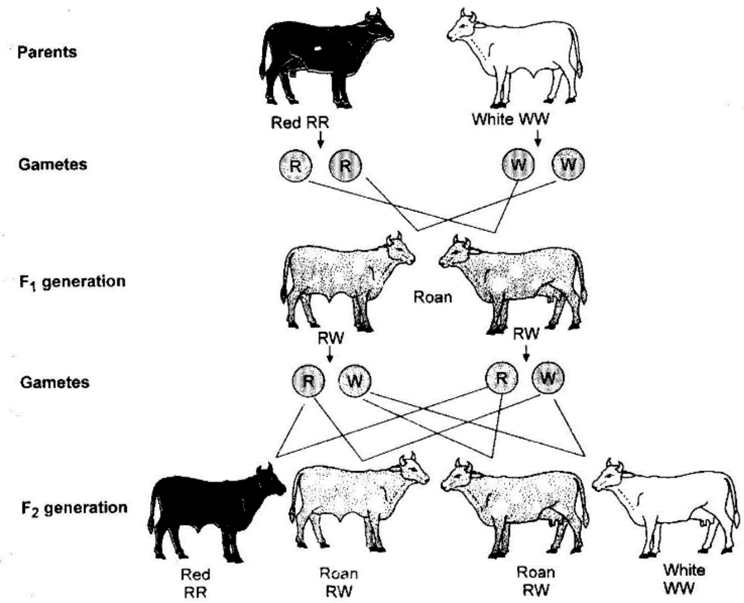
Sickle-cell anaemia is a hereditary disease. Persons suffering from sickle cell anaemia have **sickle cell haemoglobin** (Hb^S) in their RBCs. Such RBCs become sickle shaped under low oxygen concentration. In fact the disease sickle cell anaemia is caused by a gene Hb^s**.** Its normal allele is Hb^A which produces normal haemoglobin. A person with Hb^AHb^A genotype have normal haemoglobin but the persons with Hb^SHb^S have sickle cell haemoglobin, they die of lethal or fatal anaemia but heterozygous persons with Hb^AHb^S genotype have both types of haemoglobin and suffer from mild anaemia because some of their RBCs during oxygen deficiency, become sickle shaped. It shows that Hb^A and Hb^Sgenes are **co-dominant**





Example 3 - Coat colour in cattle

- In cattle gene **R** stands for red coat colour and **W** stands for white coat colour. When red cattle (RR) are crossed with white cattle (WW), hybrids of the F1 generation are of **roan** colour with genotype **RW.** The roan colour cattle comprises dark colour (red) interspersed with white colour which develop side by side in heterozygous organisms. In F2 generation red, roan and white cattle are produced in the ratio of 1 : 2 : 1 (RR : RW : WW)



Difference between Incomplete Dominance and Co-Dominance	
Incomplete Dominance	**Co-Dominance**
1. Effect of one of the two alleles is more conspicuous.	1. The effect of both the alleles is equally conspicuous.
2. It produces a fine mixture of the expression of two alleles.	2. There is no mixing of the effect of the two alleles.
3. The effect in hybrid is intermediate of the expression of the two alleles.	3. Both the alleles produce their effect independently, e.g., I^A and I^B, $Hb^s$and Hb^A.
4. The hybrid possesses a new phenotype.	4. A new phenotype does not develop.
5. The expressed new phenotype has no allele of its own	5. The expressed phenotype is combination of two phenotypes and their alleles.

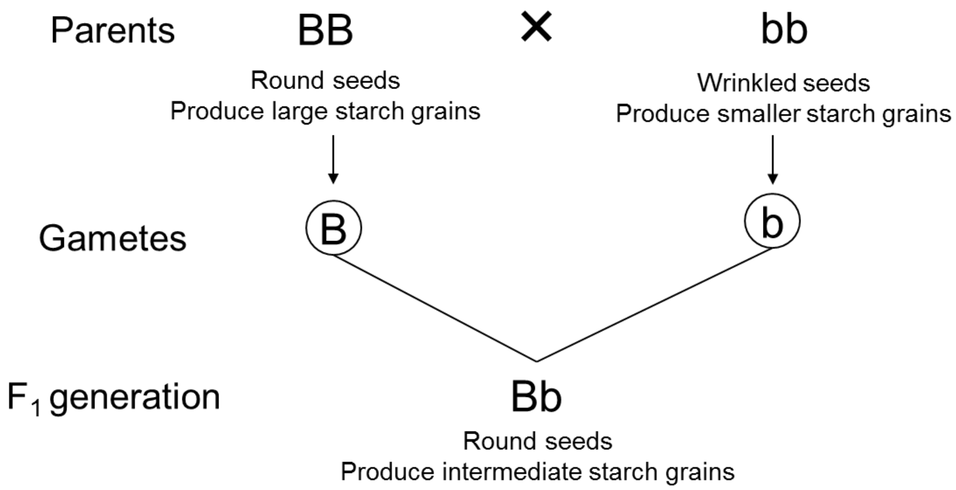
Difference between Co-Dominance and Dominance	
Co-Dominance	**Dominance**
1. Both the alleles are equally dominant.	1. Only one allele is dominant.
2. Both the alleles show their independent effects even in heterozygous condition.	2. Only one allele shows its independent effect in heterozygous condition.
3. The number of phenotypes is more than the number of alleles.	3. The number of phenotype is the same as the number of alleles.

Pleiotropy

- It is a condition in which a single gene may produce more than one trait (multiple phenotypic expression) is called **pleiotropy.** Such a gene is called **pleiotropic gene.**
- Pleiotropic gene mainly effects the metabolic pathways which contributes towards various phenotypic characters.
- Example – starch synthesis in pea seed, phenylketonuria, and sickle cell anaemia.

Example 1 – Starch synthesis in pea seeds

- Starch synthesis in pea seeds is controlled by one gene and it has two alleles **"B"** and **"b".**
- Starch is synthesized effectively by "**BB"** homozygotes and produces large starch grains.
- In contrast **"bb"** homozygotes have lesser efficiency in starch synthesis and produce smaller starch grain.
- **"BB"** seeds are round and the **"bb"** seeds are wrinkled.
- Heterozygotes produce round seeds and so **"B"** seems to be dominant allele. But the starch grain produced are of intermediate size in **"Bb"** seeds.
- So if starch grain size is considered as the phenotype, then from this angle the alleles show incomplete dominance.



NOTE –
- When a gene produce more than one phenotype then the dominance is not considered as the feature of gene or its product. It depends upon the gene product and particular phenotype we choose to examine.

Example 2 – Phenyl ketonuria (PKU)

- Discovered by **Folling** 1934
- Phenylketonuria is caused by mutation (single gene mutation – on chromosome no.12) in the gene that codes the enzyme **phenylalanine hydroxylase,** which converts the phenylalanine to tyrosine.
- Tyrosine is required for the protein synthesis and for the formation of catecholamine, thyroxine, and melanin synthesis.
- The absence of phenylalanine hydroxylase leads to conversion of phenylalanine to phenylpyruvic acid which accumulates in the brain and causes mental retardation, reduction in hair and skin pigmentation and excretion of phenylpyruvic acid through urine because of poor absorption in the kidney.

NOTE –
- It occurs in about 1 in 18000 births among white Europeans.
- It is very rare in other races.
- Blood and urine test of neonates can indicate the disease.

Example 3 – Sickle cell anaemia.

- In sickle cell anaemia, the genes which causes this disease alter the type of haemoglobin and also change the form of RBC.

Polygenic Inheritance / Multifactorial or Quantitative inheritance

- The trait which are controlled by three or more genes are called **polygenic trait** and such genes are called **polygenes** or **multiple genes** or **cumulative genes**
- In a polygenic trait the phenotype reflects the contribution of each allele, .i.e., **the effect of each allele is additive** or **cumulative**.
- Beside the involvement of multiple genes, polygenic inheritance also takes into account the influence of environment.
- Example – Human skin colour, kernel colour of wheat

Example 1 – Human skin colour
- Let us assume that the three genes **A**, **B** and **C** controls skin colour in humans.
- With the dominant form "**ABC"** responsible for **dark skin colour** and the recessive alleles **"abc"** of for **light skin colour**.
- The genotype with all the dominant alleles **(AABBCC)** will have the **darkest skin colour** and that with all the recessive alleles **(aabbcc)** will have the **lightest skin colour**.
- The genotype with three dominant alleles and three recessive alleles **(AaBbCc)** will have an **intermediate skin colour**.
- In this manner the number of each type of alleles in the genotype would determine the darkness or lightness of the skin in an individual.

NOTE –
- **F. Galton –** He studied polygenic inheritance in man and coined the term **eugenics.**
- **Eugenics –** the application of genetic in an attempt to improve the heredity qualities of humans.
- **Nilsson-Ehle –** obtained the first experimental proof of polygenic inheritance in case of kernel colour of wheat

Dihybrid Cross - Inheritance of Two Genes

- A cross between two homozygous individuals differing in two pairs of contrasting characters is called a **dihybrid cross**. (e.g., seed shape and seed colour)
- Round shape and yellow colour is dominant over wrinkled shape and green colour.
- A **homozygous pea plant** producing **round seeds** with **yellow cotyledons** designated as **(RRYY)** (where **R** represents the gene for **Round** and **Y** for **Yellow**) was **crossed** with a **homozygous pea plant** producing **wrinkled seed** with **green cotyledons** designated as **(rryy)** (where **r** represents the gene for **wrinkled** and **y** for **green**). These represents **Parent generation (P)**
- The offsprings produced in the next generation i.e., **F_1 generation** were **round seeds** with **yellow cotyledons (RrYy).**
- **Selfing of F_1 generation** plants produced **16 types** of **F_2 generation** plants. i.e., Round yellow, round green, wrinkled yellow and wrinkled green in the ration of **9 : 3 : 3 : 1.**
- Among the F_2 plants, one is with **RRYY** and one with **rryy**, where the first one is **pure dominant** and the other is **recessive**.
- The other **14 types of plants** are different from parental types due to **recombination of genes.**
- When each pair of contrasting characters are considered singly they separate independently.
- Hence, it was seen that the separation or segregation of contrasting characters of a given trait is totally independent of segregation of contrasting characters of another trait. Thus, genes for shape do not intermix with genes for colour **but separate** or **assort independently**.
- When the above dihybrid cross was worked out, the following result to the F_2 generation is obtained. It produced altogether **sixteen different combinations** as shown below in a **punnett square**.

P generation: Round Yellow RRYY × Wrinkled Green rryy

Gametes: RY, ry

F_1 generation: RrYy (all are heterozygous round yellow)

Selfing: RrYy × RrYy

Gametes: RY Ry rY ry RY Ry rY ry

F_2 generation

♂ \ ♀	RY	Ry	rY	ry
RY	RRYY (Round Yellow)	RRYy (Round Yellow)	RrYY (Round Yellow)	RrYy (Round Yellow)
Ry	RRYy (Round Yellow)	RRyy (Round Green)	RrYy (Round Yellow)	Rryy (Round Green)
rY	RrYY (Round Yellow)	RrYy (Round Yellow)	rrYY (Wrinkled Yellow)	rrYy (Wrinkled Yellow)
ry	RrYy (Round Yellow)	Rryy (Round Green)	rrYy (Wrinkled Yellow)	rryy (Wrinkled Green)

- **Phenotypic ratio – 9:3:3:1**
  i.e., 9 Round yellow : 3 Round green : 3 Wrinkled yellow : 1 Wrinkled green
- **Genotypic Ratio – 1 : 2 : 1 : 2 : 4 : 2 : 1 : 2 : 1**
  i.e., 1 RRYY : 2 RRYy : 1 RRyy : 2 RrYY : 4 RrYy : 2 Rryy : 1 rrYY : 2 rrYy : 1 rryy

NOTE –
- The ratio of 9:3:3:1 can be derived as a combination series of 3 Round : 1 Wrinkled with 3 Yellow : 1 Green. This derivation can be written as follows:
- (3 Round : 1 Wrinkled) (3 Yellow : 1 Green) = 9 Round, Yellow : 3 Wrinkled, Yellow : 3 Round, Green : 1 Wrinkled, Green

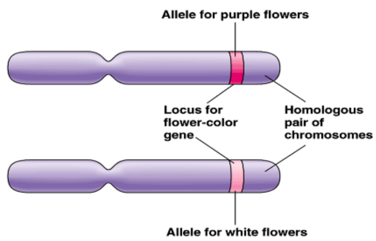
Mendel's Third Law / Law of independent Assortment

- When two pairs of traits are combined in a hybrid, one pair of trait segregates independent of the other pair of trait.
- In a dihybrid cross between two plants having round yellow (RRYY) and wrinkled green seeds (rryy), four types of gametes (RY, Ry, rY, ry) are produced. Each of these segregate independent of each other, each having a frequency of 25% of the total gametes produced.

Rediscovery of Mendel's Work
- Mendel published his work on inheritance of characters in the journal **'Proceedings of the Natural History Society'** of Brunn in 1865 but his work remained unrecognised till 1900 because of the following reasons.
  - Lack of communication and publicity
  - His concept of factors (genes) as discrete units that did not blend with each other was not accepted in the light of variations occurring continuously in nature.
  - Mendel's approach to explain biological phenomenon with the help of mathematics was also not accepted.
  - In 1900, three scientists **Hugo deVries, Correns** and **Von Tschermak** independently rediscovered Mendel's work.

Chromosomal Theory of Inheritance

- By 1900, due to the advancement in microscopy, chromosomes were also discovered.
- **Walter Sutton** and **Theodore Bovery** discovered that the behaviour of chromosomes was parallel to the behaviour of genes.
- Chromosomes and genes both occur in pairs—two alleles of a gene pair are located on **homologous sites of homologous chromosomes**.
- Sutton and Bovery further proposed that it is the pairing and separation of a pair of chromosomes that ultimately leads to segregation of the pair of factors (genes) they carry.



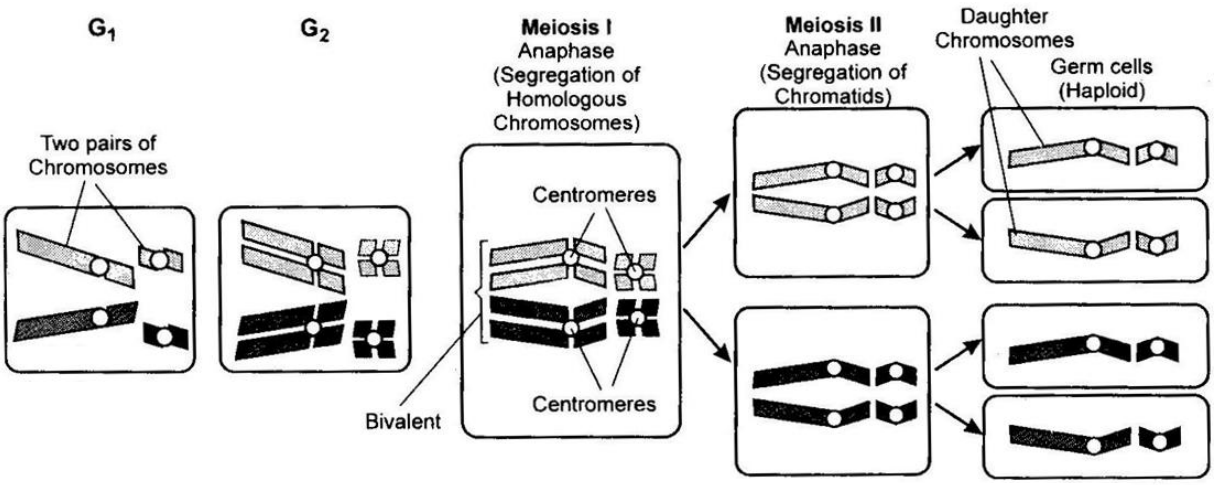


Figure: formation of four germ cells form one cell (with four chromosome) by meiosis to show segregation of homologous chromosomes.

- During anaphase of meiosis I, the two chromosomes pairs can align of the metaphase plate independently of each other. To understand this, compare the chromosomes in the left and right columns.

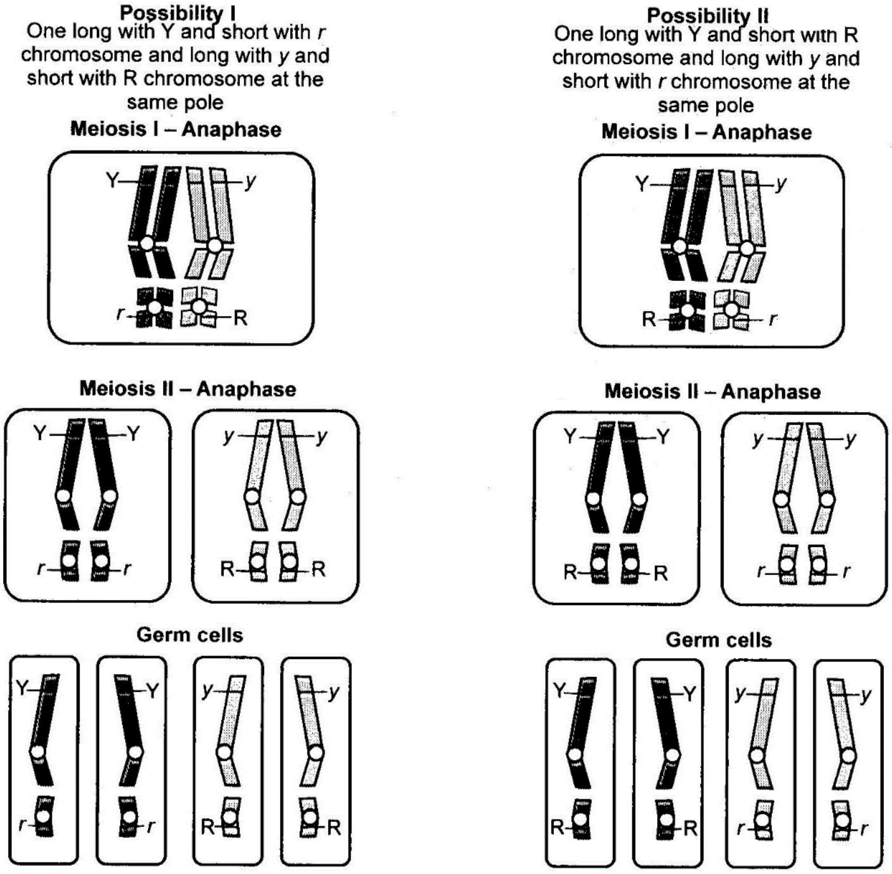


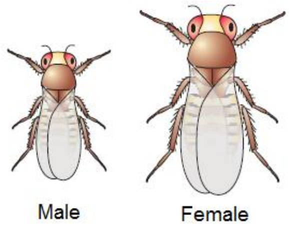
Figure: Independent assortment of chromosomes during meiosis. During Anaphase 1 of meiosis 1, the two chromosome pairs that were aligned at metaphase plate separate out randomly

- Union of knowledge of chromosomal segregation with Mendelian principles constitutes chromosomal theory of inheritance.

Comparison between the behaviour of genes and chromosomes	
GENES	**CHROMOSOMES**
Occur in pairs	Occur in pairs
Segregate at the time of gamete formation such that only one of each pair is transmitted to a gamete	Segregate at gamete formation and only one of each pair is transmitted to a gamete
Independent pairs segregate independently of each other	One pair segregates independently of another pair

Linkage and Recombination

- **Thomas Hunt Morgan ('father of experimental genetics' / 'father of sex linkage)** discovered the basis of variations that sexual reproduction produced.
- He worked on **fruit flies** (*Drosophila melanogaster*).
- He chose *Drosophila* because of the following reasons:
  - They were suitable to grow on synthetic medium in laboratory.
  - Their life cycle is complete in two weeks.
  - Single mating produces many progeny flies.
  - Clear differentiation of sexes – Easily distinguishable male and female
  - Hereditary variations clearly visible with low power microscopes



Morgan's experiment

- Dihybrid cross was carried out on fruit flies. Yellow bodied, white eyed females were crossed with brown bodied, red eyed males.
- F_1 progeny was obtained, which were inter-crossed.
- F_2 progeny was obtained and F_2 ratio was observed.
- F_2 ratio was observed to be significantly different from 9:3:3:1 as observed in Mendelian dihybrid cross.

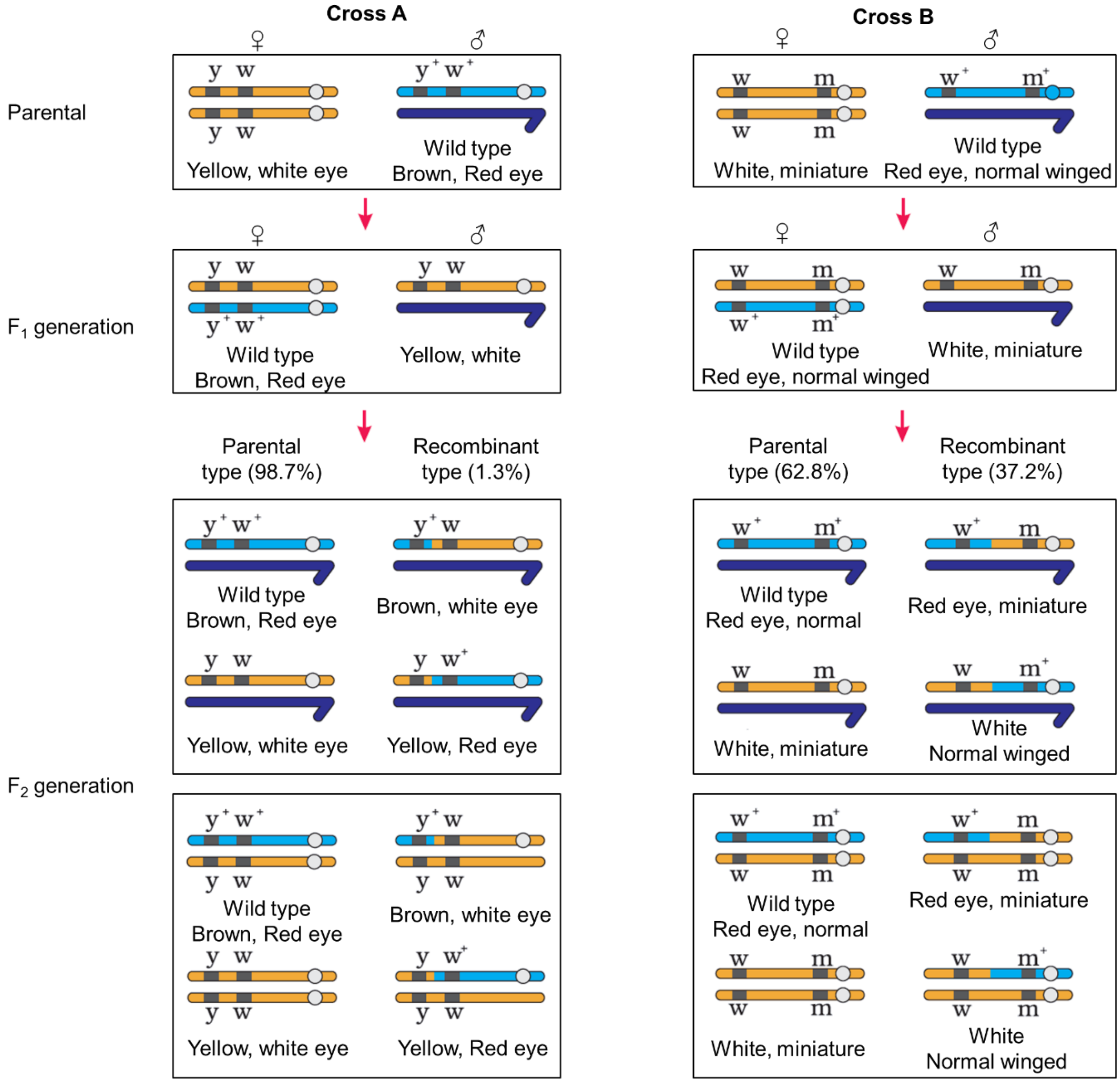


Figure 1: Incomplete linkage in Drosophila showing recombination of genes for body colour and eye colour. Results of a dihybrid cross conducted by Morgan, showing cross between gene y and w. Here dominant wild type alleles are represented with (+) sign in superscript.

Figure 2: Incomplete linkage in Drosophila showing recombination of genes for eye colour and wing size. Results of a dihybrid cross conducted by Morgan, showing cross between gene w and m. Here dominant wild type alleles are represented with (+) sign in superscript.

NOTE - The strength of linkage between y and w is higher than w and m.

Explanation of deviation from Mendelian ratio:

- Genes involved are located on X chromosome.
- When two genes are located on the same chromosome, the proportions of parental gene combinations were much higher than those of non-parental.
- **Linkage –** Physical association of genes on a chromosome
- **Recombination –** Non-parental gene combination

Recombination

Low Recombination

- Occurs when genes on same chromosome are tightly linked
- Example – genes white and yellow are tightly linked and show only 1.3% recombination.

High Recombination

- Occurs when genes on same chromosome are loosely linked
- Example – Genes white and miniature are loosely linked and show 37.2% recombination.

- **Alfred Sturtevant** utilized the knowledge of frequency of gene recombination as a measure of physical distance between two genes and to map their position on chromosomes.
- In this way, genetic maps were prepared, which are extensively used today for genome sequencing projects as in human genome project.

NOTE –

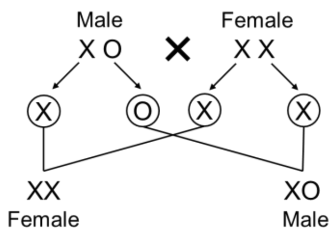
- The distance between genes is measured by **map unit (.m.u).** 1% crossing over between two linked genes is known as **1 map unit** or **Centi Morgan (cM).** 10% crossing over as **deci Morgan (dM)** and 100% crossing over is termed as **Morgan (M)**.

- The physical distance between two genes determines both the strength of the linkage and the frequency of the crossing over between two genes. The strength of the linkage increases with the closeness of the two genes. On the other hand the frequency of the crossing over increases with the increase in the physical distance between the two genes.

Sex Determination

- **Henking** discovered the genetic/chromosomal basis of sex determination by working on insects. He observed specific nuclear structures during spermatogenesis in insects. He named these structures as **X bodies.**
- He observed that after spermatogenesis, 50% of the sperm obtained these structures, while 50% did not.
- Later on, it was found that the X body observed by Henking was actually a chromosome and thus, this chromosome was named **X chromosome**.
- Chromosomes involved in sex determination are called **allosomes (sex chromosomes)**, while the other chromosomes are called **autosomes**.

XO type of sex determination

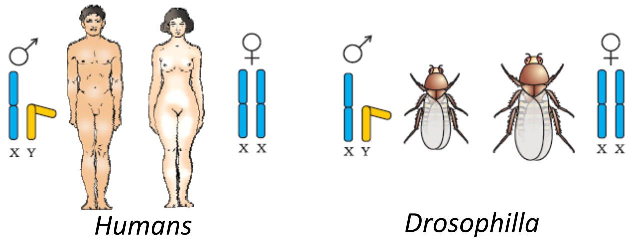


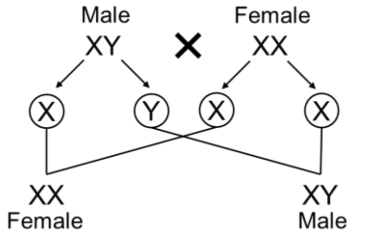
- Other than autosomes, at least one X chromosome is present in all insects.
- Some sperms contain X chromosomes, while some do not.
- Eggs fertilised by sperms having X chromosomes become females. So, females have two X chromosomes.
- Eggs fertilised by sperms not having X chromosomes become males. So, males have only one X chromosome.
- Example of organisms with XO type of sex determination – Grasshopper, Earthworm

XY type of sex determination

- The chromosome pattern in the human female is XX and that in the male is XY.
- All the haploid gametes produced by the female (ova) have the sex chromosome X.
- Males produce 50% of sperms carry the X chromosome while the other 50% would carry the Y.

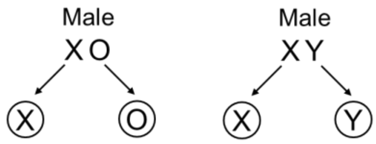
- After fertilisation the zygote would carry either XX or XY depending on whether the sperm carrying X or Y fertilized the ovum.
- The zygote carrying XX would develop into a female baby and XY would form a male
- Example of organisms with XY type of sex determination – Humans and *Drosophila*





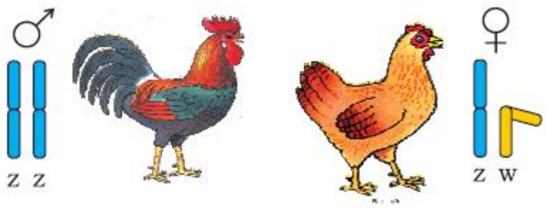
Male heterogamety

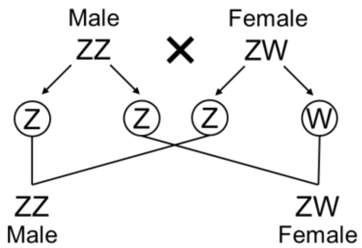
- XO and XY types of sex determination are examples of male heterogamety.
- In XO type, some gametes have X chromosomes, while some gametes are without X chromosomes.
- In XY type, some gametes have X chromosomes, while some gametes have Y chromosomes.



Female heterogamety

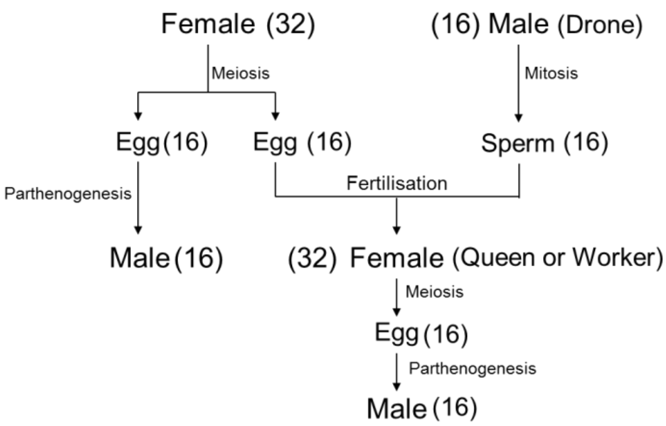
- ZW type of sex determination is an example of female heterogamety.
- In ZW type, the female has one Z and one W chromosome, while the male has a pair of Z chromosomes. Example – Birds and some reptiles and fishes





Haplodiploid Sex-Determination in Honey Bee

- The sex determination in honey bee is based on the number of sets of chromosomes an individual receives.
- An offspring formed from the union of a sperm and an egg develops as a female (queen or worker), and an unfertilized egg develops as a male (drone) by means of parthenogenesis.
- This means that the males have half the number of chromosomes than that of a female. The females are diploid having **32 chromosomes** and males are haploid, i.e., having **16 chromosomes**. This is called as **Haplodiploid sex-determination system.**
- **Males produce sperms by mitosis, they do not have father and thus cannot have sons, but have a grandfather and can have grandsons**.



Mutation

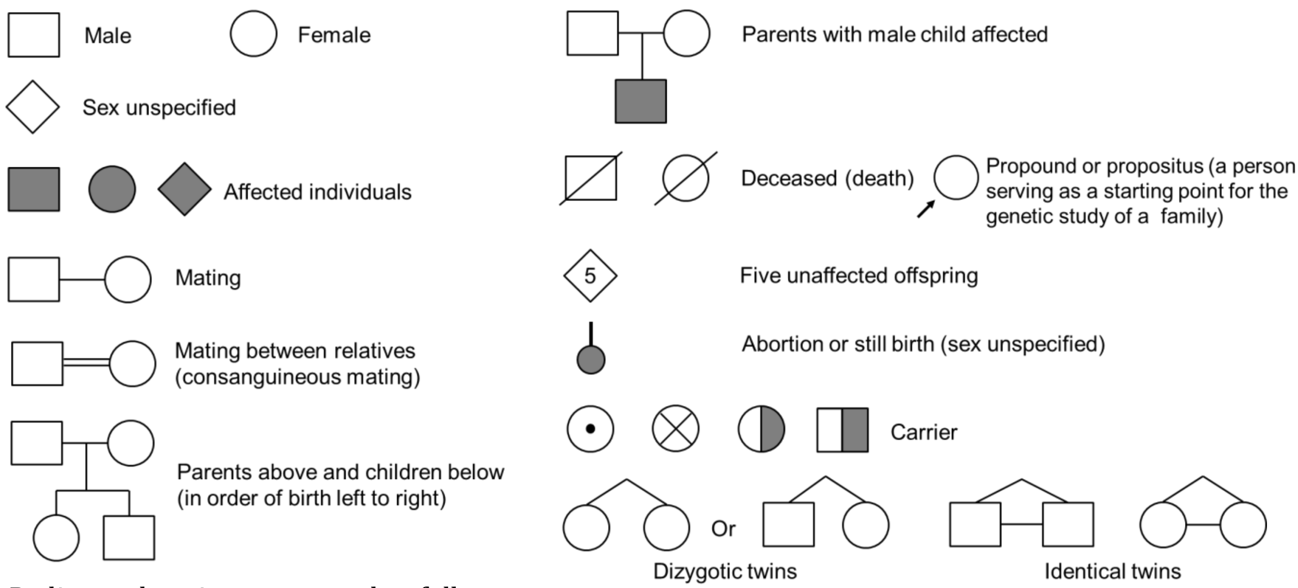
- Alteration of DNA sequence resulting in changes in genotype and phenotype of organisms is called **mutation**.
- **Mutagens** – are the substance which causes mutation. Following are the examples.
    - Physical mutagens – UV radiations, Ionizing radiations, Gamma rays.
    - Chemical mutagens – Nitrous acid (deaminating agent which changes cytosine to uracil).
    - Biological mutagens – viruses (e.g. *HPV*), bacteria (e.g. *Helicobacter*)
- DNA helix runs in a chromatid, hence any change (insertion or deletion) in the DNA sequence affects the chromosome.

Types of mutation

- **Point Mutation** – Mutation arising due to change in single base pair of DNA as in sickle cell anaemia
- **Frameshift Mutation** – Mutations arising due to deletion or insertion of base pair in DNA sequence

Pedigree Analysis

- Pedigree analysis is the analysis of inheritance of traits in several generations of a family.
- A particular trait under study is represented in a family tree.
- By using pedigree analysis, inheritance of a specific trait, abnormality or disease, can be traced.
- DNA is believed to be the carrier of genetic information, which passes unaltered from generation to generation. Mutations occasionally alter the genetic material and genetic diseases are believed to be associated with these alterations only.
- Standard symbols in pedigree analysis are as follows:



- Pedigree chart is represented as follows:

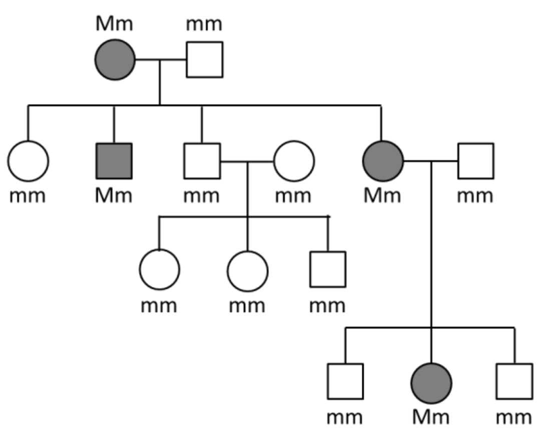


Figure: (a) Autosomal dominant trait (For example: Myotonic dystrophy)

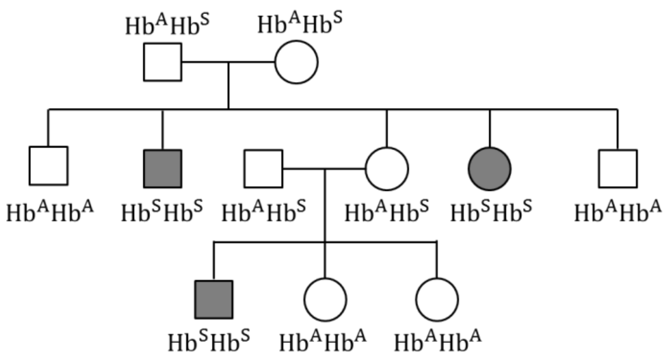


Figure: (b) Autosomal recessive trait (For example: Sickle-cell anaemia)

Genetic Disorders

- Include Mendelian disorders and chromosomal disorders

Mendelian Disorders

- Characterized by **mutation in a single gene**.
- Their mode of inheritance follows the principles of Mendelian genetics.
- Mendelian disorders can be

  1. **Autosomal dominant trait**
  2. **Autosomal recessive trait**
  3. **Sex linked trait**

1. Autosomal dominant trait

- These are the trait whose encoding gene is present on any of the autosomes and the **wild type allele is recessive** to its mutant allele, i.e., the **mutant allele is dominant**.
- Examples – Muscular or Myotonic dystrophy, Huntington disease (HD) (ch4), Brachydactyly, Polydactyly, Dimple in the cheek, Widows peak, Tongue rolling, Tasting of PTC (phenylthiocarbamide/Phenylthiourea)

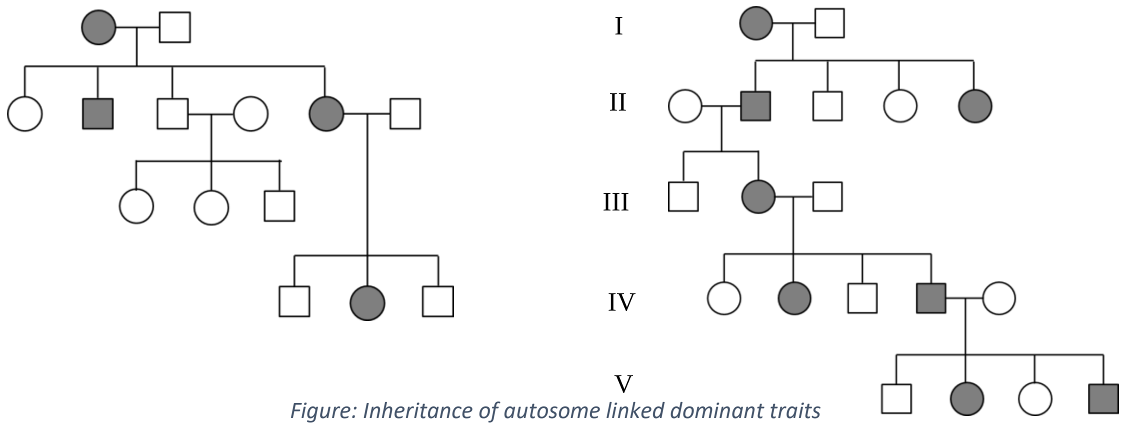


Figure: Inheritance of autosome linked dominant traits

The characteristic features of inheritance of autosome linked dominant traits are:

- Transmission of traits occurs from parents of either sex.
- Males and females are equally affected.
- The pedigree is vertical, i.e., the trait is marked to be present in each of the generations.
- Multiple generations are characteristically affected.

2. Autosomal recessive trait

- These are the traits whose mutant allele is recessive to its wild type allele.
- Examples – Sickle cell anaemia (ch11), Thalassemia (ch16 and Ch11), Phenylketonuria (ch12), Cystic fibrosis (ch7), Albinism, Fused ear lobe, Alkaptonuria (black urine disease), Gaucher's disease (ch1), Tay-Sach's diseases (TSD) or Infantile amourotic idiocy or GM2 gangliosidosis.

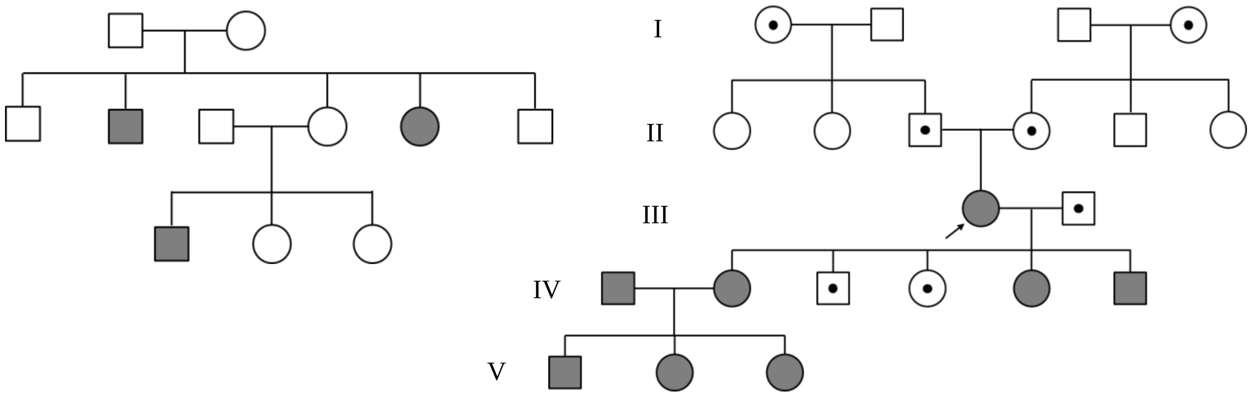


Figure: Inheritance of autosome linked recessive traits

The characteristic features of inheritance of autosome linked recessive traits are:

- Occur in equal proportion in multiple male and female siblings, whose parents are normal but carriers.
- The siblings are homozygous for the defective allele, but their parents, though some may appear normal, are obviously heterozygous, i.e., are merely carriers of the trait.
- Consanguinity (marriage between man and woman genetically related to each other, such as cousins) occasionally results in the appearance of such traits.

Sickle cell anaemia

- Autosomal recessive disease
- Transmission – From parent to offspring when both parents are carriers of disease
- Pair of alleles Hb^A and Hb^S controls the expression of this disease.
  Hb^A and Hb^A – Normal
  Hb^A and Hb^S – Carrier of disease
  Hb^S and Hb^S – Diseased
- **Cause of the disease** – Change in gene at **chromosome no.11** causes the replacement of **GAG** by **GUG** leading to the substitution of **Glu** by **Val** at **6th** position of **beta globin chain** of haemoglobin.
- The mutant haemoglobin so formed polymerises at low oxygen tension, resulting in change in shape of RBC to sickle-like.

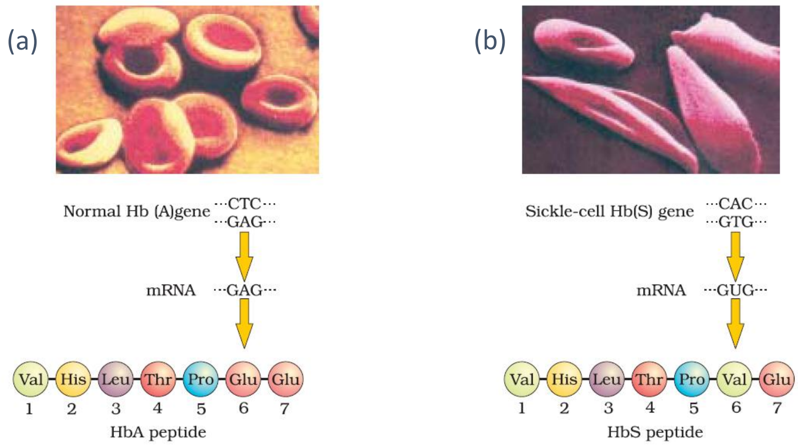
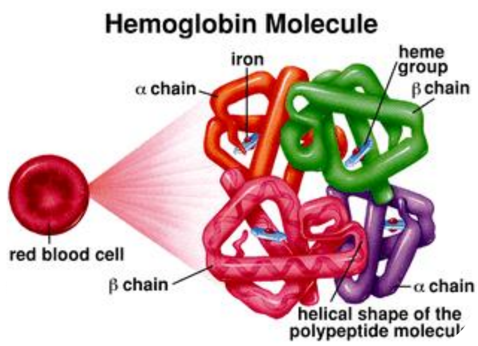


Figure: micrograph of the red blood cells and the amino acid composition of the relevant portion of β-chain of haemoglobin: (a) From a normal individual: (b) From an individual with sickle-cell anaemia

Thalassemia (Gr. Thalassa – Sea i.e., Mediterranean; Anaemia – weak blood)

- Autosomal linked recessive blood disease
- Transmission - from parents to the offspring when both the partners are unaffected carrier for the gene (heterozygous)
- Body makes an abnormal form of haemoglobin (quantitative)
- This results in excessive destruction of RBC which leads to anaemia
- Haemoglobin is made of two polypeptide chains namely Alpha globin chain and Beta globin chain.
- Thalassemia occurs when there is a defective gene or genes for any one of the chain



Types of Thalassemia

- Based on the type of polypeptide chain affected

  i) Alpha thalassemia
  ii) Beta thalassemia

i) Alpha thalassemia

- Production of alpha globin chain is affected.
- Alpha globin chains are encoded by two closely linked gene (**HBA1** & **HBA2**) on **chromosome 16**
- Occurs in people from Southeast Asia, Middle East and China.

ii) Beta thalassemia

- Production of Beta globin chain is affected.
- Beta globin chains are encoded by a single gene (**HBB**) on **chromosome 11** of each parent.
- Occurs due to mutation in one or both the gene

Phenylketonuria (PKU)

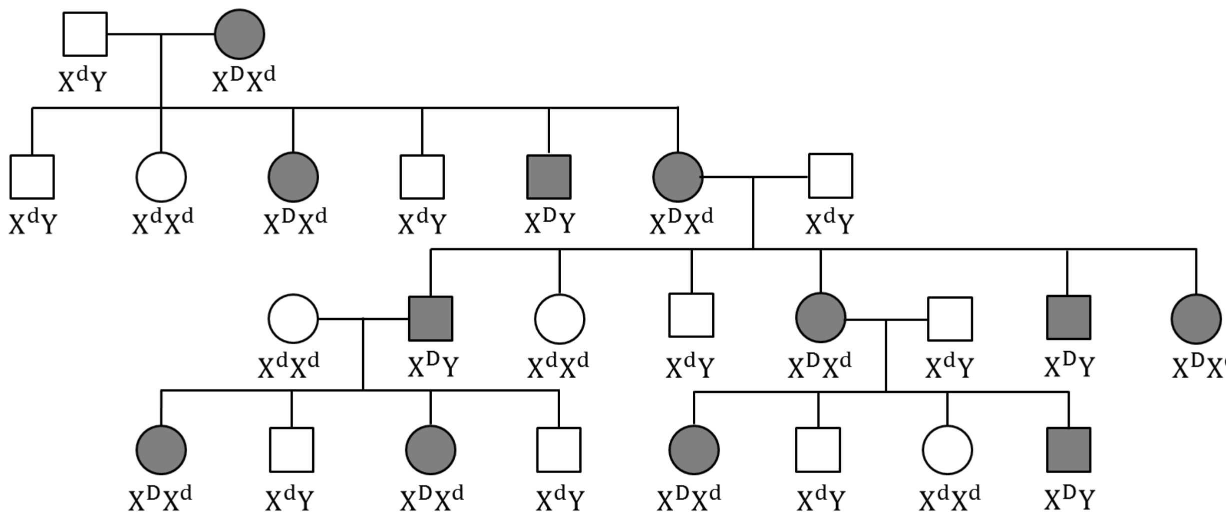
- Discovered by **Folling** 1934
- Phenylketonuria is caused by mutation (single gene mutation – on **chromosome no.12**) in the gene that codes the enzyme **phenylalanine hydroxylase,** which converts the phenylalanine to tyrosine.
- Tyrosine is required for the protein synthesis and for the formation of catecholamine, thyroxine, and melanin synthesis.
- The absence of phenylalanine hydroxylase leads to conversion of phenylalanine to phenylpyruvic acid which accumulates in the brain and causes mental retardation, reduction in hair and skin pigmentation and excretion of phenylpyruvic acid through urine because of poor absorption in the kidney.

3. Sex linked trait

- X-linked dominant trait - Duchenne muscular dystrophy
- X-linked recessive trait - Haemophilia, Colour blindness
- Y-linked trait – Hypertrichosis

X-linked dominant trait

- These are the traits whose encoding gene is present on the X – chromosome and the mutant allele of which is dominant over its wild type allele
- Example – **Duchenne muscular dystrophy.**
- Here, the dominant mutant allele is denoted by ‘D’, and its recessive wild type allele is denoted by‘d’. Remember that human females have two X-chromosomes (XX) and the males have only one X and one Y chromosome. Males receive their lone X – chromosome from their mother and the Y – chromosome from their father, whereas females receives one of her X – chromosome form her mother and the other X from her father.

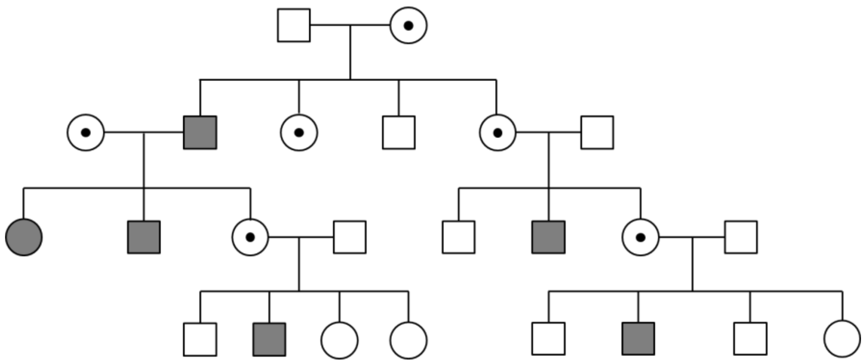


The characteristics of inheritance of X-linked dominant trait are:

- The trait appears in almost all the generations and the inheritance is vertical.
- If the female is affected, then about half of her sons and daughters are affected.
- If the male is affected then all of his daughters would be affected, but none of his sons are affected.
- In short, the pedigree resembles the pattern of inheritance of autosomal dominants, except that there is no male-to-male transmission in X-linked dominant.

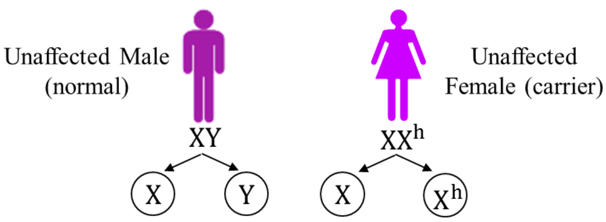
X-linked recessive trait

- These are the trait whose encoding gene is present on the X-chromosome and its mutant allele is recessive to its wild-type allele.
- Examples – **Haemophilia**, **Red-green colour blindness (Daltonism)**.

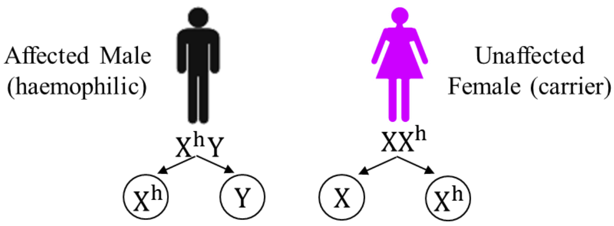


The characteristics features of inheritance of X-linked recessive trait are:
- Female express the trait only when they are homozygous for the mutant allele, whereas the male do so even when they are **hemizygous** for it.
- About half of the sons of the carrier (heterozygous for the trait) females are affected. In case of homozygous females showing the trait, all of her sons are likely to be affected. Therefore, the males are most affected in the population.
- Affected persons are related to one another through the maternal side of their family.
- Any evidence of male-to male transmission of the trait rules out the X-linked inheritance.

Example 1 - Haemophilia A (Royal disease/Bleeders disease)
- Haemophilia is a X-linked recessive disease
- In this disease, protein (factor VIII) involved in blood clotting is affected. Therefore, even a simple cut results in uncontrolled bleeding.
- Transmission – From unaffected female (carrier) to male progeny
- Females act as carriers of disease, but rarely suffer from haemophilia since for a female to become haemophilic, the mother should be carrier and father should be haemophilic.



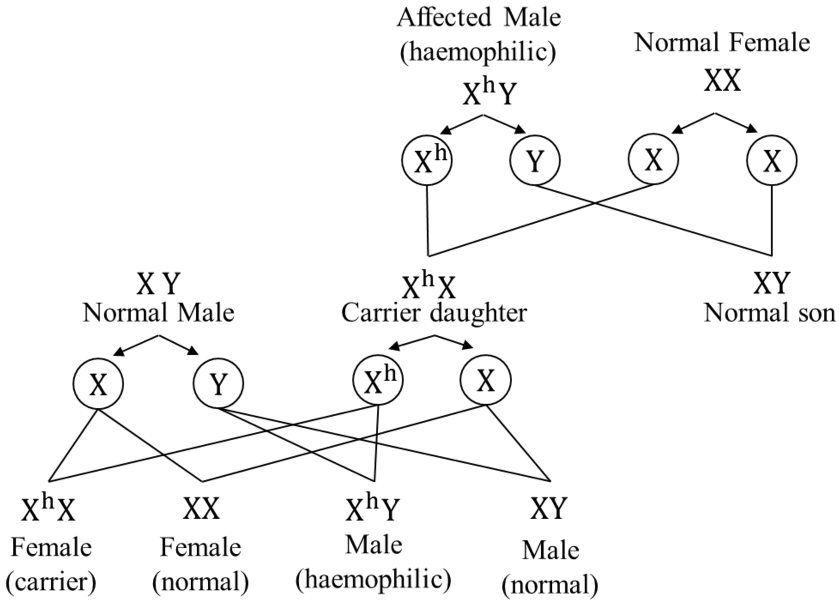
♂ \ ♀	X	X^h
X	XX Unaffected female (normal)	X^hX Unaffected Female (carrier)
Y	XY Unaffected Male (normal)	X^hY Affected Male (haemophilic)



♂ \ ♀	X	X^h
X^h	XX^h Unaffected female (carrier)	X^hX^h Affected female (haemophilic)
Y	XY Unaffected male (normal)	X^hY Affected male (haemophilic)

Extra information

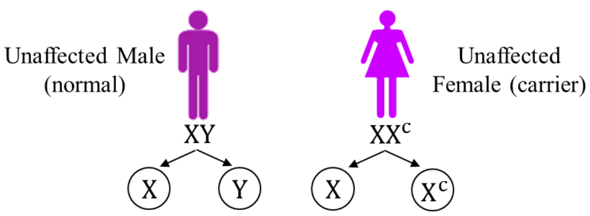
- X-linked recessive trait shows **criss-cross / zig zag inheritance.** i.e., the grandson inherit the trait from his grandfather through his mother who is carrier of the gene for trait.



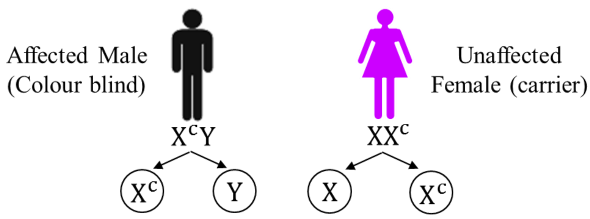
- Haemophilia is also referred to as the **'royal disease'** because several members of royal families in Europe were affected by it.
- An example for this is **Queen Victoria.**

Example 2 - Colour Blindness (red-green colour blindness – Daltonism)

- Colour blindness is a X-linked recessive disorder due to defect in either red or green cone of eye resulting in failure to discriminate between red and green colour.
- This defect is due to mutation in certain genes present in the X chromosome.
- It occurs in about **8% of males** and only about **0.4% of females** (because females have two X chromosomes).
- The son of woman who carriers the gene has a 50% chance of being colour blind. The mother is not herself colour bind because the gene is recessive. That means that its effect is suppressed by her matching dominant normal gene.
- A daughter will not normally be colour blind, unless her mother is carrier and her father is colour blind.
- All females will definitely be a colour blind, if both of her parents are also colour blind.



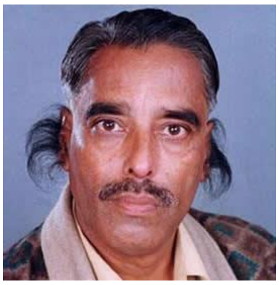
♂ \ ♀	X	X^c
X	XX Unaffected female (normal)	X^cX Unaffected Female (carrier)
Y	XY Unaffected Male (normal)	X^cY Affected Male (colour blind)

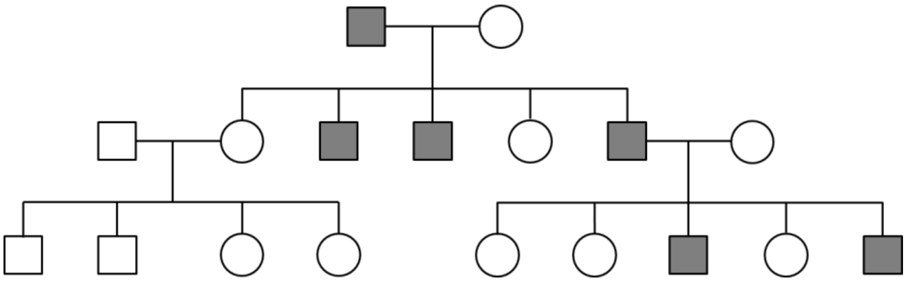


♂ \ ♀	X	X^c
X^c	XX^c Unaffected female (carrier)	X^cX^c Affected Female (colour blind)
Y	XY Unaffected Male (normal)	X^cY Affected Male (colour blind)

Y-linked trait (male-sex limited trait)

- These are the traits whose gene is present on the Y-chromosome.
- The females do not have any Y-chromosome.
- All the males must have a Y-chromosome which is obtained from their father
- Any trait linked the Y-chromosome must be present only in males and certainly not in any of the females. Hence these traits are also called as **male-sex limited trait**.
- All the sons of the affected male would express the trait whereas none of his daughters would do so.
- Example – **Hypertrichosis** (presence of hairs on pinna)
- The pattern of the pedigree chart would be as follows.



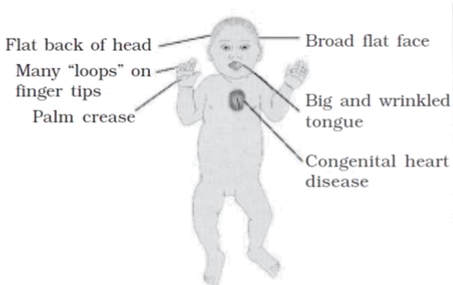


Chromosomal Disorders

- Chromosomal disorders are disorders characterized by **changes in the chromosome number** and **structure** which causes a series of abnormalities collectively known as **syndrome.**
- They may either effect the **autosomes** or the **allosomes**
- Total number of chromosomes in humans = 46 (23 pairs)
- Total 23 pairs = 22 pairs of autosomes (body chromosomes) + 1 pair of allosomes (sex chromosomes)
- **Aneuploidy –** Loss or gain of chromosomes due to failure of segregation of chromatids during cell division.
- **Monosomy (2n-1) –** Lack of one chromosome in any one pair of homologous chromosomes.
- **Trisomy (2n+1)–** Inclusion of an additional copy of chromosomes in any one pair of homologous chromosomes.

Down's syndrome (Mongolism)

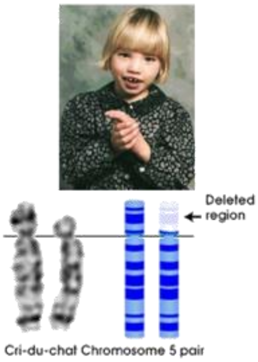
- First described by **Langdon Down** (1866)
- **Cause:** presence of an additional copy of chromosome 21 (**autosomal trisomy of 21**)
- **Symptoms:** short stature, small, round head, furrowed tongue, partially opened mouth, palm crease, congenital heart disease and mental retardation.



Cri du chat syndrome ("cry of the cat")

- **Cause:** is a rare syndrome caused by a **deletion of the short arm of chromosome 5** or **partial Monosomy** during the development of an egg or sperm.
- This syndrome derives its name from French which means **"cry of the cat"** referring to the infant's cry which is high pitched and sounds like a cat due to abnormal larynx development
- **Occurrence** – it is a rare disease and occurs at the rate of one in 50,000 live births.

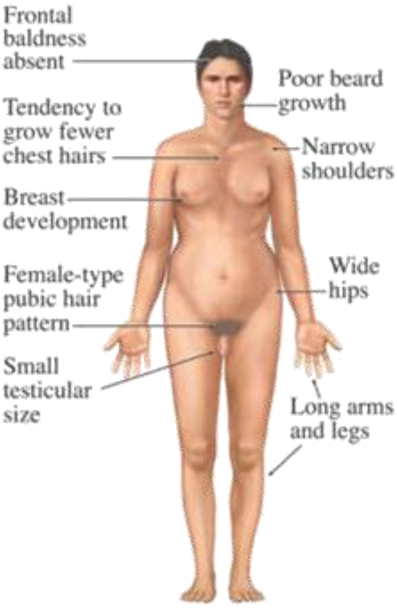
Symptoms



- Produce high pitched sound which sounds like cry of the cat
- Infants have low birth weight
- May have respiratory problem
- Severe mental retardation
- Slow and incomplete development of motor skills
- Some people with this disorder have a shortened lifespan, but most have normal life expectancy

Klinefelter's Syndrome

- **Cause:** Additional copy of X chromosome, i.e., **47 chromosomes (XXY) (allosomal trisomy)**
- **Symptoms:** individual is sterile with an overall masculine development with gynaecomastia



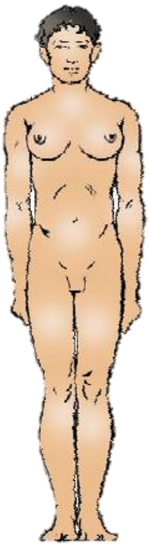
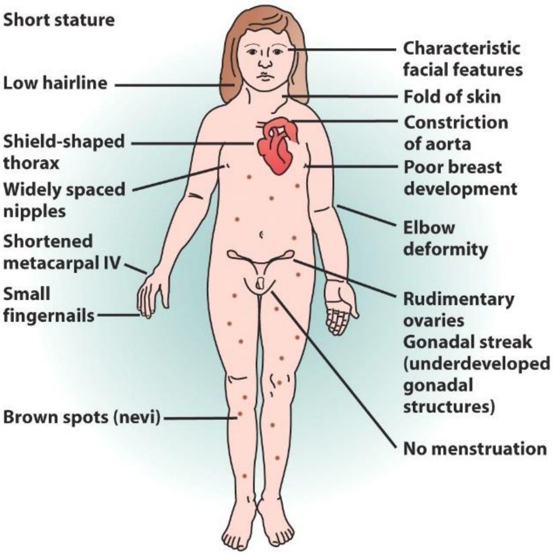


Figure: Tall stature with feminized characters

Turner's Syndrome

- **Cause:** Absence of one X chromosome, i.e., **45 chromosomes (XO) (allosomal monosomy)**
- **Symptoms:** Affected females are sterile; have rudimentary ovaries; secondary sexual characters are absent



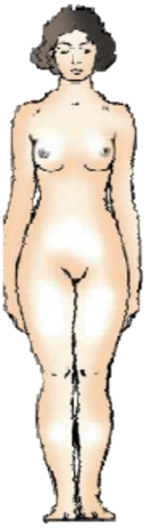


Figure: Short stature and underdeveloped feminine character

SUMMARY

Genetics is a branch of biology which deals with principles of inheritance and its practices. Progeny resembling the parents in morphological and physiological features has attracted the attention of many biologists. Mendel was the first to study this phenomenon systematically. While studying the pattern of inheritance in pea plants of contrasting characters, Mendel proposed the principles of inheritance, which are today referred to as 'Mendel's Laws of Inheritance'. He proposed that the 'factors' (later named as genes) regulating the characters are found in pairs known as alleles. He observed that the expression of the characters in the offspring follow a definite pattern in different–first generations (F_1), second (F_2) and so on. Some characters are dominant over others. The dominant characters are expressed when factors are in heterozygous condition (Law of Dominance). The recessive characters are only expressed in homozygous conditions. The characters never blend in heterozygous condition. A recessive character that was not expressed in heterozygous condition may expressed again when it becomes homozygous. Hence, characters segregate while formation of gametes (Law of Segregation).

Not all characters show true dominance. Some characters show incomplete, and some show co-dominance. When Mendel studied the inheritance of two characters together, it was found that the factors independently assort and combine in all permutations and combinations (Law of Independent Assortment). Different combinations of gametes are theoretically represented in a square tabular form known as 'Punnett Square'. The factors (now known as gene) on chromosomes regulating the characters are called the genotype and the physical expression of the characters is called phenotype.

After knowing that the genes are located on the chromosomes, a good correlation was drawn between Mendel's laws: segregation and assortment of chromosomes during meiosis. The Mendel's laws were extended in the form of 'Chromosomal Theory of Inheritance'. Later, it was found that Mendel's law of independent assortment does not hold true for the genes that were located on the same chromosomes. These genes were called as 'linked genes'. Closely located genes assorted together, and distantly located genes, due to recombination, assorted independently. Linkage maps, therefore, corresponded to arrangement of genes on a chromosome.

Many genes were linked to sexes also, and called as sex-linked genes. The two sexes (male and female) were found to have a set of chromosomes which were common, and another set which was different. The chromosomes which were different in two sexes were named as sex chromosomes. The remaining set was named as autosomes. In humans, a normal female has 22 pairs of autosomes and a pair of sex chromosomes (XX). A male has 22 pairs of autosomes and a pair of sex chromosome as XY. In chicken, sex chromosomes in male are ZZ, and in females are ZW.

Mutation is defined as change in the genetic material. A point mutation is a change of a single base pair in DNA. Sickle-cell anaemia is caused due to change of one base in the gene coding for beta-chain of haemoglobin. Inheritable mutations can be studied by generating a pedigree of a family. Some mutations involve changes in whole set of chromosomes (polyploidy) or change in a subset of chromosome number (aneuploidy). This helped in understanding the mutational basis of genetic disorders. Down's syndrome is due to trisomy of chromosome 21, where there is an extra copy of chromosome 21 and consequently the total number of chromosome becomes 47. In Turner's syndrome, one X chromosome is missing and the sex chromosome is as XO, and in Klinefelter's syndrome, the condition is XXY. These can be easily studied by analysis of Karyotypes.